MyFit Tracker - Complete Setup Guide

An installation, configuration and usage guide for MyFit Tracker, a fitness tracking project created by Vishnu Prasad.

Read this first. Seven rules that protect your data.

1. Never commit `.env.local` or any credential to Git.
2. Never share your GitHub token, and never paste it into chat, screenshots or issues.
3. Never make your fitness-data repository public. Keep it private.
4. Never put a secret in a client-side variable. Anything starting with `NEXT_PUBLIC_` is shown to every visitor.
5. A deployment URL is not an access-control mechanism. Anyone who finds the URL can open the pages.
6. Before you store sensitive data, verify which routes are protected (Chapter 17 explains what is and is not).
7. Keep regular backups, and do not share screenshots or exports that contain private data.

Free hosting plans and service limits may change. Always check the current terms of GitHub and Vercel.

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1. Introduction and project overview

MyFit Tracker is a mobile-first web app for tracking workouts, food, water and body measurements. It is built with Next.js and React.

How your data is handled:

- **Local first.** Everything you enter is saved in your browser on your device straight away. The app works without any server setup.
- **Optional cloud sync.** If you configure it, you can press *Sync now* to copy your data to a **private GitHub repository that you own**. A small server route (`/api/data`) does this using a token that never reaches your browser.
- **No tracking.** The app contains no analytics or telemetry. Its only network traffic is to its own `/api/data` route, which talks to GitHub from the server.

What the app includes: a Home dashboard, a repeating seven-day training cycle with set logging, a Leave button to skip a day, a food log with your own macros, water quick-add, weight and waist logs, progress charts, a weekly summary, Excel export, JSON backup and restore.

How the pieces fit together:

1. **Source repository** (the code). You fork or clone it. It contains no personal data.
2. **Data repository** (your records). A separate **private** repo that only you control.
3. **Hosting** (optional). Vercel runs the app so you can use it from your phone.

2. Prerequisites

- A computer with **Node.js 20 or newer** and **npm** (npm comes with Node.js). Download Node.js from nodejs.org and choose the LTS version.
- **Git**, to download and upload code.
- A **GitHub account** (free is fine).
- A **Vercel account** if you want to host it online (optional; you can run it locally).
- A code editor such as VS Code (optional but helpful).

Check your tools by running these in a terminal:

```
node -v
npm -v
git --version
```

3. Creating or accessing a GitHub account

1. Go to github.com and choose *Sign up*, or *Sign in* if you already have an account.
2. Verify your email address.
3. Turn on two-factor authentication: GitHub, then Settings, then *Password and authentication*. This protects the account that holds your tokens and data.

4. Forking or cloning the source repository

The canonical source is https://github.com/greynoddigital/myfit-tracker.

Option A: fork (simple, keeps a link to upstream).

1. Open the repository and choose *Fork*, then *Create fork*.
2. Your copy appears under your account. Note that a fork of a public repository is itself public. That is fine because the source contains no secrets, but never put secrets in it.

Option B: private copy. Clone the repository, create a new **private** empty repository on GitHub, then push to it (see Chapter 10).

Download the code to your computer (replace `YOUR-USERNAME`):

```
git clone https://github.com/YOUR-USERNAME/myfit-tracker.git
cd myfit-tracker
npm ci
```

`npm ci` installs the exact dependency versions recorded in `package-lock.json`.

> **Licence note.** At the time this guide was written, the project had not yet chosen an open-source licence. Check the repository's `README.md` and `LICENSE` file for the current terms before you reuse or redistribute the code.

5. Creating the separate private fitness-data repository

This repository stores your personal records as JSON files. It must be private.

1. On GitHub choose *New repository*.
2. Name it, for example, `myfit-data`.
3. Set visibility to **Private**.
4. Tick *Add a README file*. This matters: the branch must already exist before the app can write to it.
5. Choose *Create repository*.

The app will create a `data/` folder in this repo on the first sync, with files such as `data/workouts.json` and `data/food.json`. You do not create these by hand.

> **Warning:** never make this repository public and never invite people you do not trust. It contains your health-related records.

6. Creating a fine-grained GitHub personal access token

The app needs a token so the server can read and write only your data repository. Create a **fine-grained** token (not a classic token).

1. On GitHub open your profile menu, then *Settings*, then *Developer settings*, then *Personal access tokens*, then *Fine-grained tokens*, then *Generate new token*. (Menu names can change slightly over time.)
2. **Token name:** `myfit-sync`.
3. **Expiration:** choose a limited lifetime, for example 90 days, and add a reminder to renew it.
4. **Resource owner:** your own account.

5. **Repository access:** choose *Only select repositories* and select **only** your data repository (for example `myfit-data`). Do not choose *All repositories*.
6. **Permissions:** under *Repository permissions* set **Contents** to **Read and write**. *Metadata: Read-only* is added automatically. Nothing else is needed, because the app only reads and writes files through the repository contents API.
7. Choose *Generate token* and copy the value now. GitHub shows it only once.

Store the token only in `.env.local` and in Vercel's environment variables. Never in Git, chat, screenshots or notes that sync to the cloud.

7. Configuring local environment variables

In the project folder, copy the example file:

```
cp .env.example .env.local
```

On Windows PowerShell use `Copy-Item .env.example .env.local`.

Open `.env.local` in an editor. The file `.env.local` is already listed in `.gitignore`, so Git will not upload it. Still, double-check before every commit (Chapter 10).

8. Configuring APP_PASSWORD, GITHUB_TOKEN, GITHUB_REPO and GITHUB_BRANCH

Fill in the four values exactly as follows:

Variable	What to enter
`GITHUB_TOKEN`	The fine-grained token from Chapter 6. Secret.
`GITHUB_REPO`	Your data repo as `owner/name`, for example `your-username/myfit-data`. This is the data repo, not the source repo.
`GITHUB_BRANCH`	The branch in the data repo. Use `main` unless yours is different. If left empty the app uses `main`.
`APP_PASSWORD`	A long, unique passphrase that you invent (for example four or more random words). Secret.

Example shape (use your own values, never copy these):

```
GITHUB_TOKEN=your-fine-grained-token-here
GITHUB_REPO=your-username/myfit-data
GITHUB_BRANCH=main
APP_PASSWORD=a-long-unique-passphrase-you-chose
```

Important facts about `APP_PASSWORD`:

- If it is empty or missing, the server refuses **every** sync request.
- It is one shared secret. There is no rate limiting or lockout, so a short password can be guessed. Make it long and unique.
- Never prefix any of these four names with `NEXT_PUBLIC_`.

Two optional **public** variables, `NEXT_PUBLIC_OPERATOR_NAME` and `NEXT_PUBLIC_OPERATOR_URL`, let you identify yourself as the operator of your own deployment in the footer. They are visible to every visitor.

9. Running the application locally

```
npm run dev
```

Open http://localhost:3000 in your browser. After you change `.env.local`, stop the server (Ctrl+C) and start it again.

Test cloud sync:

1. In the app open *Settings*, then *Cloud Sync* (the *Sync and app password* row).
2. Type your `APP_PASSWORD` in the *App password* field. It is saved in this browser automatically.
3. Press *Sync now*.
4. You should see *Synced*. Open your data repo on GitHub: a `data/` folder with JSON files should appear.

Sync is **manual**. Nothing is uploaded until you press *Sync now*. Press it after you finish logging, and before you switch devices.

To check the production build locally: `npm run build` then `npm start`.

10. Pushing customized source code to GitHub

If you changed code or want your own copy online:

```
git status
git add .
git commit -m "Describe your change"
git push origin main
```

Before every commit run `git status` and confirm that **no** `.env.local`, backup, export or personal file is listed. If a secret was ever committed, treat it as exposed: revoke the token in GitHub right away and create a new one. Deleting the file in a later commit does not remove it from Git history.

For a **private copy** of the source (Option B in Chapter 4):

```
git remote rename origin upstream
git remote add origin https://github.com/YOUR-USERNAME/YOUR-PRIVATE-REPO.git
git push -u origin main
```

11. Importing the project into Vercel

1. Go to vercel.com and sign in with your GitHub account.
2. Choose *Add New*, then *Project*.
3. Under *Import Git Repository* select your source repository. If it is not listed, choose the option to adjust GitHub permissions and grant Vercel access to that repository only.
4. Vercel detects **Next.js** automatically. Leave the build settings at their defaults.
5. **Stop before deploying** and add the environment variables (next chapter).

12. Configuring environment variables in Vercel

On the import screen, open *Environment Variables* and add all four names and values from Chapter 8: `GITHUB_TOKEN`, `GITHUB_REPO`, `GITHUB_BRANCH`, `APP_PASSWORD`.

- Do not add a `NEXT_PUBLIC_` prefix to any of them.
- Mark the secrets as sensitive if Vercel offers that option.
- To change a variable later, edit it under *Project, Settings, Environment Variables* and **redeploy**. Changes do not apply to an existing deployment.

13. Deploying and verifying the application

1. Choose *Deploy* and wait for the build to finish.
2. Open the URL Vercel gives you on your phone and computer.
3. Verify sync: *Settings, Cloud Sync*, enter the app password, press *Sync now*. Expect *Synced*.
4. Verify that the password is enforced: enter a wrong password and press *Sync now*. The app shows *Sync failed*, which means the server refused the request (it returned HTTP 401). Then re-enter the correct password.
5. Check your data repo on GitHub for new commits named `sync ...`.

On a phone you can add the site to your home screen from the browser menu for quick access.

Remember: **the URL is public.** The pages load for anyone who has the address; the password protects only the data route. Vercel offers additional deployment-protection options on some plans; check Vercel's current documentation and pricing.

14. Using the tracker

Home. Shows today's workout, the week strip, your nutrition overview, daily protein and creatine check-ins, and the weekly check-in prompt.

Workouts and the seven-day cycle. The plan repeats every seven days: Day 1 Hybrid Push, Day 2 Hybrid Pull, Day 3 Quad-Focused Legs, Day 4 Rest, Day 5 Upper Body, Day 6 Glute and Hamstring-Focused Legs, Day 7 Rest, then Day 1 again. With the default settings Day 1 falls on a Monday, so the two rest days are Thursday and Sunday until you press Leave. To start a session, open *Workouts* and press *Start Workout*; log weight, reps and notes for each set and finish the workout. The app shows what you lifted last time.

Leave. If you cannot train today, press **Leave** on Home or Workouts and confirm. Today is marked **Skipped** (never Completed) and every later workout and rest day moves forward one day, keeping their order. Cancel changes nothing. Leave is not offered on rest days, on a day already skipped, or once a workout has been started.

Editing the plan. *Settings, Weekly Training Plan* lets you change which exercises are on each cycle day (tabs D1 to D7). *Settings, Manage Gym Equipment* lets you rename equipment and add exercises.

Food. Open *Food*, press *Add Food* or *Quick Add*, pick a meal (Breakfast, Lunch, Dinner, Snacks), and enter calories, protein, carbs and fat yourself. The app has no built-in food database or barcode scanner. You can save foods as favourites, duplicate an entry, edit a portion, and move between days. Set daily targets under *Settings, Goals and Targets*.

Water. Use the quick-add buttons (250, 500 or 1000 ml) or the one-tap +250 ml on the Food page. Set your daily water target under Goals and Targets.

Body and progress. Use *Settings, Weekly check-in*, *Weight log* and *Waist log* to record measurements, and *Progress* to see strength, weight, waist and water trends.

Reports, exports and backups (all under *Settings*):

- *Weekly summary*: review a week and press *Export weekly report* to download a Markdown file.
- *Export to Excel*: last 7 days, a custom range, or all history, as an `.xlsx` file.
- *Backup, Download JSON*: saves everything as `MyFit_Backup.json`.
- *Import Backup, Restore from JSON*: replaces **all** current data after a confirmation.
- *Reset All Data*: needs you to type RESET. If sync is on, the next sync overwrites the cloud copy as well, so download a backup first.

Exports and backups contain your personal records. Store them privately and never commit them.

15. Updating the app and pulling upstream changes

Your fork can receive improvements from the original project.

```
git remote add upstream https://github.com/greynoddigital/myfit-tracker.git
git fetch upstream
git merge upstream/main
npm ci
npm test
npm run build
git push origin main
```

If `git remote add upstream` says the remote exists, skip that line. Resolve any merge conflicts in your editor, then run the checks. Vercel redeploys automatically when you push. **Download a JSON backup before updating.**

16. Troubleshooting common errors

Problem	What to check
Sync failed	The server could not complete the request. Check that the token is valid and not expired, that `GITHUB_REPO` is spelled `owner/name`, that the branch exists (the data repo needs at least one commit, so initialise it with a README), and that the token has *Contents: Read and write* on that repo.
Sync failed after you changed or retyped the password	The app shows the same *Sync failed* message for a wrong password. Check first that the password typed in *Settings, Cloud Sync* matches `APP_PASSWORD` exactly, and that `APP_PASSWORD` is set on the server (a missing value makes the server refuse everything).
Remote has newer data	Another device synced first. Sync from the device that has the newest data. Use *Keep my data and overwrite* only if you are sure your copy is the right one.
Changes to environment variables have no effect	Locally, restart `npm run dev`. On Vercel, redeploy.
`npm ci` fails	Install Node.js 20 or newer and try again.
Vercel build fails	Open the build log. Run `npm run build` locally to reproduce.
Data missing on a new phone or after clearing the browser	Data is stored per browser. On a new device enter the app password and press *Sync now* to pull your data, or use *Import Backup*.

Problem	What to check
Sync keeps failing although the token and password are right	Make sure the branch named in `GITHUB_BRANCH` really exists in the data repo (initialise the repo with a README). Also confirm the token has not expired.

17. Security, privacy, backups and recovery

What was verified in the source code:

- The GitHub token is used only on the server (`lib/github.ts`, called by `app/api/data/route.ts`). No `NEXT_PUBLIC_` variable carries a secret.
- The data route requires the `x-app-password` header to equal `APP_PASSWORD`, otherwise it returns 401. It only reads and writes a fixed list of file names inside the one repository named by `GITHUB_REPO`.
- Stale writes are rejected so that newer cloud data is not silently overwritten.
- Import and export work in the browser and upload nothing by themselves.
- There is no analytics or tracking.

Limits you must understand:

- The site address is public; only the data route is password-protected.
- The password has no rate limiting. Use a long, unique one.
- The password you type is kept in your browser's local storage on that device, and local data is not encrypted. Do not use shared or public computers.
- Anyone with the password can read and overwrite all synced data.

Backups and recovery:

1. Download a JSON backup regularly (*Settings, Data Management, Backup*) and keep it somewhere private.
2. Your private data repo keeps the Git history of every sync, so earlier versions of each `data/*.json` file can be viewed on GitHub.
3. To restore, use *Import Backup* with a JSON file, or sync from a device that has good data.
4. If a token may be exposed: revoke it on GitHub (Settings, Developer settings, Fine-grained tokens), create a new one, and update `.env.local` and Vercel. If `APP_PASSWORD` may be exposed: choose a new one, update it in Vercel, redeploy, and re-enter it in the app.
5. Never share screenshots or exports containing private records.

18. Creator attribution, licence and contribution guidance

MyFit Tracker was designed and built by **Vishnu Prasad**. The canonical repository is https://github.com/greynoddigital/myfit-tracker.

- **Independent deployments.** You may run your own copy. Your deployment is yours: it is not necessarily owned, maintained or endorsed by the original author. You can show your name as operator in the footer with `NEXT_PUBLIC_OPERATOR_NAME` and `NEXT_PUBLIC_OPERATOR_URL`. See `docs/ATTRIBUTION_AND_FORKS.md`.
- **Keep attribution accurate.** Do not claim you created the original project, and keep the original credit and licence notice as the licence requires.
- **Licence.** At the time of writing no licence had been selected. Check the repository's `LICENSE` file. See `docs/LICENSE_RECOMMENDATION.md`.

- **Contributing.** Improvements are welcome upstream. Read `CONTRIBUTING.md`, run `npx tsc --noEmit`, `npm test` and `npm run build`, and never include secrets or personal data. Report security issues privately as described in `SECURITY.md`.

19. Thank you and official links

Thank you for using MyFit Tracker. Built for healthier habits. Shared with everyone.

Official links:

- Vishnu Prasad (personal website): https://vxnuprasad.github.io/
- GreyNod Labs (cybersecurity and security testing): https://greynodlabs.com
- GreyNod Digital (web development and digital services): https://greynoddigital.github.io
- Original MyFit Tracker repository: https://github.com/greynoddigital/myfit-tracker